



PCB ID Folder — Updated Sitemap

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152

PCB FOLDERS

142

TEMPLATE NODES

49

PLACEHOLDER SLOTS

378

DFM/DFT CHECKS

Rev-0...N

MISSING LAYER

G0-G3

DFX GATES

The canonical template `1. Product ID_Template Folder` defines five areas across 142 nodes — but 49 of its 72 files are `blank file.docx` placeholders, and it has no concept of revisions. Every genuinely worked folder invents a revision layer of its own: `Eb-21-EL-287-01-1628-GW-109` runs to Rev-7 across Gerber, ODB, Step, Schematic, Project Files, BOM and Pick-and-Place. This structure adopts what real work already does.

1 · Proposed structure

<PCB-ID>/	e.g. Eb-21-EL-287-01-1628-GW-109
├ 00-Audit-Checklist.xlsx	exit artefact · one tab per board
├ 01-Input-from-Client/	ONE spelling – three variants exist today
├ 02-Hardware/	
│ └ 00-Design/	
│ │ Block-Diagram/ · Schematic/ · Layout/ · Simulation/	
│ │ Design-Review/	gate evidence, named approver
│ │ Documentation/	HDD · datasheet · design notes · power profiling
│ └ 01-Outputs/	THE REVISION LAYER – NEW
│ │ └ Rev-0/ · Rev-1/ · ... · Rev-N/	
│ │ │ Gerber/ ODB/ Netlist/ Step/ Schematic-PDF/ 3D-PDF/	
│ │ │ Project-Files/ BOM/ Pick-and-Place/	
│ │ └ CURRENT-REVISION.txt	names the live Rev – one line, no ambiguity
├ 02-Testing/	
│ Electrical/ · Environmental/ · Functional/ · Thermal/ · Burn-In/	
│ └ each: Parameters/ · Procedure/ · Report/	
└ Defect-Log/	NEW
├ 03-Firmware/	
│ └ 00-Design/	Block-Diagram/ State-Machine/ Flowchart/ Logic-Sheet/ Architecture/
│ └ 01-Source/	repo link + release tags – not loose zips
│ └ 02-Builds/	Rev-N/ signed binary + git hash NEW
└ 03-Testing/	Unit/ Static-Analysis/ On-Target/ Integration/ HIL/ OTA/ Soak/ Defect-Log/
├ 04-Enclosure/	
│ └ 00-Design/	Concept/ CAD/ Renders/ 2D-Drawings/ Tolerance-Stack/
│ └ 01-Outputs/	Rev-N/ → Step/ DXF/ Manufacturing-Drawings/ Mechanical-BOM/ Vendor-SAR/ NEW
└ 02-Testing/	Material/ Drop/ IP/ Vibration/ Mechanism-Life/ Cosmetic-FAI/ Defect-Log/
├ 05-DFx-Manufacturability-and-Test/	NEW – the gate that blocks fab release
│ └ 00-Standards-and-Capability/	customer rule deck + the fab and EMS capability sheets
│ └ 01-Master-Checklist/	EB-T-023 – 378 checks, 72 parameters, gated G0-G3
│ └ 02-Reviews-DFS-DFC-DFE/	service · cost · environment
│ └ 03-Action-Register-and-Waivers/	every finding, owner, date, disposition
│ └ 04-Gate-Sign-off/	four signatures per gate – Fab · EMS · Production · QA
│ └ 05-FMEA/	DFMEA · PFMEA
│ └ 06-Test-Hooks-and-Provisioning/	diagnostic command set · programming and provisioning plan
├ 06-Assembly/	

Hardware-Assembly/ stencil · stack-up · pick-and-place · wiring · line doc
 Product-Assembly/ drawings · sequence · wire harness · mechanical BOM

07-System-Test/**NEW — nothing exists for this today**

Integration/ · EMC-Pre-Scan/ · Pre-Compliance/ · Compliance-Lab/
 Certificates/ cert + the build revision it covers
 HALT-HASS/ · MTBF/ · Field-Trial/ · UAT-Sign-Off/

08-Builds/**NEW — build identity**

EVT/ · DVT/ · PVT/
 each: build record · BOM used · yield · issue log

09-Delivery/**NEW**

Delivery-Tracking-Sheet · handover pack · code docket

DELETE: root **Testing/** folder — empty in all three live folders inspected.
 Testing happens inside Hardware Testing and Firmware Testing.

2 · The DFM / DFT gate — new

DFx was never a place. It was nine steps scattered through hardware design, one in enclosure and four in mass production, with no folder, no owner and — until this revision — two templates between them. The 308-check DFM/DFT checklist that Elecbits actually built, traced to PollEx, the SE rule deck, IEC 60664-1, IEC 60884-1, IEC 62368-1 and IPC-2221, was sitting in an archive folder. It is now the master, generalised off the smart plug and extended to 378 checks.

GATE	CLOSES AT	CHECKS	WHAT IT PROTECTS
G0	Before layout begins	60 active	Design input. The governing standard, the fabricator and EMS capability sheets, and the agreed parameter set. The cheapest gate in the process — everything it catches costs nothing to fix, everything it misses costs a respin.
G1	After DFx review 2	147 active	The layout itself. Board and panel, placement, pad and stencil, routing, drill, silkscreen, assembly poka-yoke, thermal. The bulk of the file, answered against the board that is actually drawn.
G2	After DFx sign-off, before fab release	64 active	Buildability and testability. Test point provision, geometry, four-wire measurement, coverage and production test. Shared with the EMS through the Customer Extract view; internal rows stay internal.
G3	Mass production, before release	67 active	The product and the line. Production safety testing, coating, marking, compliance and FMEA closure — the things a working prototype never proves.

Design for Test was the real hole. Test points were decided at schematic time and never re-verified after placement and routing moved them. The only later question — "do the fixtures give the coverage the parameters require?" — was asked at Mass Production, roughly 200 steps too late to change the board. Four DFT steps now run on the finished layout: test point coverage analysis, probe access and fixture feasibility, programming and debug access, and disposition of every net that cannot be reached. The Net Accessibility register carries the coverage number, and it is verified again against the real fixture before production release.

TEMPLATE	NAME	ROLE
EB-T-023	DFM/DFT Master Checklist	378 checks · 72 parameters · 7 tabs. Setup tab switches whole sections on and off by product module, so a battery product and a mains product run the same master. Carries the net accessibility register, the waiver register and a live gate dashboard.

TEMPLATE	NAME	ROLE
EB-T-214	Fabricator and EMS Process Capability Sheet	The supplier's real limits, confirmed in writing. Feeds the Parameters tab.
EB-T-215	DFx Action Item Register	Every finding with an owner, a date and a disposition. Closes the action-closure step.
EB-T-216	DFx Gate Sign-off Sheet	Four signatures per gate — Fab, EMS, Production, QA — against a named file revision.
EB-T-217/218/219	DFS · DFC · DFE Reviews	Service, cost and environment. Three steps that existed with nothing behind them.
EB-T-222/223	DFMEA · PFMEA	Design and process failure modes, with severity never reduced by an action.
EB-T-224/225	Diagnostic Command Set · Programming and Provisioning Plan	The firmware side of DFT. A command for every functional test parameter, and the order of operations that turns a blank board into a locked, serialised unit.

3 · The revision model

This is the single highest-value addition. It is not new practice — it is codifying what engineers already do informally.

RULE	DETAIL
Rev-N per output set	Gerber, ODB, Netlist, Step, Schematic PDF, 3D PDF, Project Files, BOM and Pick-and-Place all move together. A revision is a coherent set, never one file.
Never overwrite	A released revision is immutable. A change creates Rev-N+1; the old revision stays for the design history.
CURRENT-REVISION.txt	One line naming the live revision and the date it was released. Removes the "which folder is current" question entirely.
Revision on the checklist	Every Links entry in the Audit Checklist names the revision it points at, so evidence is tied to a configuration.
Certificates cite a revision	Certification attaches to a configuration, not a product name. Without the revision recorded, a later component change cannot be assessed.

4 · Audit Checklist integration

The checklist is the exit artefact for this folder. It sits at the root, carries one tab per board, and every step in the process that produces an artefact here writes four cells against its row: **Status**, **Link**, **Last updated** and **Updated by**. A row with a link but no status is an artefact nobody has accepted; a status with no link is a claim with no evidence.

CHANGE TO THE CHECKLIST	REASON
Add <code>Step ID</code> column	Without it a step's completion cannot be tested, because rows are artefact-shaped and steps are not.
Add <code>Updated by</code> column	Nothing is attributable today. For the control of record, this is the first gap to close.
Lock <code>Status</code> and <code>Approval Status</code> to dropdowns	Nine spellings of status are in use, including a free-text entry reading "Drop, Vibration, Thermal - SE has infra for this".
Add a Gates & Decisions section	Design reviews and gate releases produce a verdict, not a file, and currently have nowhere to land.

CHANGE TO THE CHECKLIST	REASON
Make it tab-per-board by default	GW-108 has three tabs — Repeater, LED, HMI — duplicated by hand because the template assumes one board.
Scope rows at design freeze	Mark each Required / Conditional / Not-Applicable so completion percentage means something. GW-122 reads 100% filled and is 77 bulk "Not started" entries with four real links.

5 · What the live folders prove

FOLDER	FOLDERS	FILES	SIZE	COVERAGE OF THE FIVE TEMPLATE AREAS
Eb-21-EL-287-01-1628-GW-109	192	583	360 MB	All five present and populated. Rev-0 to Rev-7 history. The only one with real firmware source and a delivery tracking sheet.
Eb-21-EL-287-01-1579-GW-108	126	84	27.9 MB	Hardware, Firmware and Assembly populated. Enclosure hollow — template BOM and EDD only.
IP53-YF55-XB49-SS-43	58	25	16.7 MB	Folders exist, contents largely absent. Enclosure, Firmware and Assembly empty. 34 empty folders.

The template under-specifies real work. GW-109 grew a revision layer, a delivery tracking sheet, and two differently-spelled client-input folders because the template gave it none of them. Meanwhile the template ships a root Testing folder that is empty in every live folder inspected.

6 · Naming

ITEM	FORMAT	EXAMPLE
Artefact	[PCB-ID]_<Name>_Rev<N>	Eb-21-EL-287-01-1628-GW-109_Gerber_Rev3
Document	[PCB-ID]_<Name>_v<X.Y>	..GW-109_HDD_v1.2
Dated record	[PCB-ID]_<Name>_[YYYY-MM-DD]	..GW-109_Defect-Log_2026-08-16
Board ID	<CHIP>-<PERIPHERALS>-<GW SS>-<seq>	ES3C3-JNIC-LS05-RELY5-C1CT5-GW-115
Client-linked board	<ProjectID>-<GW SS>-<seq>	Eb-21-EL-287-01-1628-GW-109

Resolve the ID conflict first. 138 of 152 Eb-Hardware folders follow the component-string convention and 20 carry a client project-ID prefix — two schemes in one folder. Casing drifts across EB- , Eb- and eb- , and two folders carry trailing whitespace. No naming rule can be enforced until one scheme is chosen.

Based on a full-depth crawl of the template folder (142 nodes), three live PCB folders (1,068 entries) and all 14 audit checklists in Eb-Hardware parsed cell by cell. Structure aligned to the 305-step Master Process Flow.